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## METHOD FOR PRODUCING WIG AND A WIG

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### Abstract

A method for producing wig. The method includes: preparing a first wig component obtained through a machine hair implantation process; dividing an elastic wig cap into different regions; stitching and securing the first wig component to the first region of the elastic wig cap; hand-tied a second wig component on the second region of the elastic wig cap; trimming a transitional area between the first region and the second region to generate a wig, etc.

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### Background/Summary

#### TECHNICAL FIELD

[0001] The invention relates to wig product, and in particular relates to a method for producing wig and a wig.

#### BACKGROUND

[0002] Wig products are highly popular as they can be worn and secured to the wearer's head, thereby transforming their overall hairstyle and look.

[0003] Traditionally, wig can be primarily divided into two types based on their manufacturing processes: machine-made wigs and hand-tied wigs. Hand-tied wigs are obtained by manually crocheting individual strands of hair onto a cap with small holes. This production method relies on manual labor, requiring a significant amount of time and incurring high labor costs. Additionally, the knots formed during the crocheting process can be visible, making the wig easily detectable when viewed up close. Therefore, the advantages of the hand-tied wigs are light and go upwards, and the disadvantages of the hand-tied wigs are expensive and too much time consuming.

[0004] Machine-made wigs, on the other hand, are produced using high post bed machine to produce the wigs. This method offers higher production efficiency and significantly reduces overall production costs from traditional hand-tied wigs. However, the machine-made wigs is much heavier than the hand-tied wig and the hair of the machine-made wigs is goes downwards.

[0005] Therefore, there is an urgent need to provide wig products that balance production costs and wearing comfort to meet the increasing demand for wigs.

## SUMMARY

[0006] In first aspect, the present application is provided a method for producing wig. The method comprises: preparing a first wig component obtained through a machine hair implantation process; dividing an elastic wig cap into a first region and a second region; stitching and securing the first wig component to the first region of the elastic wig cap; hand-tied a second wig component on the second region of the elastic wig cap; trimming a transitional area between the first region and the second region to generate a wig.

[0007] Optionally, the step of preparing a first wig component obtained through a machine hair implantation process comprises: fixing hair bundles with a clamp; wherein the hair bundles are formed by converging multiple hair strands; creating a simulated scalp consists of a layered arrangement of a first scalp layer, a second scalp layer, and an elastic mesh; using a hair implantation machine to pass a root of the hair bundles through the first scalp layer of the simulated scalp; wherein the roots of the hair bundles emerge from the first scalp layer of the simulated scalp, while the ends of the hair bundles are positioned on the outer side of the elastic mesh; removing the first scalp layer of the simulated scalp; adhering and securing the roots of the hair bundles onto the second scalp layer to generate the first wig component.

[0008] Optionally, before the step of removing the first scalp layer of the simulated scalp, the method further comprises: using a wide-toothed comb to arrange the roots of the hair bundles on the first surface of the simulated scalp; pruning the roots of the hair bundles to a predetermined length.

[0009] Optionally, wherein before the step of adhering and securing the roots of the hair bundles onto the second scalp layer, the method further comprises: pruning the roots of the hair bundles again to the predetermined length; wherein the predetermined length is the length at which the hair strands protrude from the second scalp layer.

[0010] Optionally, the predetermined length is 2 mm.

[0011] Optionally, the first scalp layer is a simulated scalp made of PVC material, and the second scalp layer is a simulated scalp made of PU material, with the second scalp layer significantly thinner than the first scalp layer.

[0012] Optionally, the step of adhering and securing the roots of the hair bundles onto the second scalp layer to generate the first wig component comprises: coating adhesive on the mating surface of the second scalp layer and placing it in an oven for baking; wherein the mating surface is the surface where the roots of the hair bundles protrude from the second scalp layer; after baking, reapplying adhesive on the mating surface of the second scalp layer and attaching a mesh layer; allowing the attached mesh layer on the simulated scalp to air dry naturally to prepare the first wig component.

[0013] Optionally, the first wig component is stitched and secured to the elastic wig cap using a transparent thread.

[0014] Optionally, the hair strands are made of synthetic fiber, human hair or mixture of both.

[0015] Optionally, the frontal part of the elastic wig cap is a curved first elastic lace net, and the remaining part of the elastic wig cap is an unshaped second elastic lace net; wherein the first elastic lace net and the second elastic lace net are stitched together using transparent thread.

[0016] In second aspect, the present application is provided a wig obtained by the method for producing wig described above.

[0017] At least one advantageous aspect of the wig and its preparation method provided in this application embodiment is that the elastic lace net is divided into two separate regions for implanted area and tied area. This approach improves upon the drawbacks of slow speed and high cost associated with fully hand-tied wigs. Meanwhile, the first wig component obtained by implantation machine process is lighter and more comfortable than the traditional machine-made wigs.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0018] One or more embodiments are described by way of example in conjunction with the accompanying drawings, which are not intended to limit the embodiments. Elements with the same reference numerals in the drawings represent similar components, unless otherwise specified. The drawings are not to scale.

[0019] FIG. 1 illustrates the flowchart of a method for producing the wig according to an embodiment of the present application.

[0020] FIG. 2 illustrates the flowchart of the method for producing the first wig component according to an embodiment of the present application.

[0021] FIG. 3 illustrates a schematic diagram of the structure simulated scalp of the according to an embodiment of the present application.

[0022] FIG. 4 illustrates a schematic diagram of the component obtained after machine implantation according to the embodiment of the present application.

[0023] FIG. 5 illustrates a schematic diagram of trimming the component shown in FIG. 4.

[0024] FIG. 6 illustrates a schematic diagram of removing the first scalp layer from the component shown in FIG. 5.

[0025] FIG. 7 illustrates a schematic diagram of applying adhesive and fixing the hair roots on the component shown in FIG. 6.

[0026] FIG. 8 illustrates a schematic diagram of the adhesive layer and the attached mesh on the component shown in FIG. 7.

[0027] FIG. 9 illustrates a schematic diagram of the elastic wig cap according to an embodiment of the present application.

[0028] FIG. 10 illustrates a schematic diagram of the hand tied process according to an embodiment of the present application.

[0029] FIG. 11 illustrates a comparison diagram between the traditional machine-made product and the first wig component obtained by the machine implanted techniques according to an embodiment of the present application.

### DETAILED DESCRIPTION

[0030] To facilitate a better understanding of the present application, a more detailed description is provided below, in conjunction with the accompanying drawings and specific embodiments. It should be noted that when an element is described as “fixed to” another element, it can be directly on the other element, or there may be one or more intermediate elements between them. When an

element is described as “connected” to another element, it can be directly connected to the other element, or there may be one or more intermediate elements between them. The positional terms used in this specification, such as “up,” “down,” “inside,” “outside,” “vertical,” “horizontal,” etc., are based on the orientation or positional relationship shown in the drawings for the purpose of describing and simplifying the present application, and should not be construed as limiting the devices or elements to a specific orientation, construction, or operation. Therefore, they should not be understood as limiting the present application.

[0031] Furthermore, terms such as “first,” “second,” etc., are used for descriptive purposes only and should not be construed as indicating or implying relative importance. Terms such as “comprising” or “including” imply the presence of the elements or objects listed before the term and do not exclude other elements or objects. The terms “connected” or “coupled” are not limited to physical or mechanical connections but can also include electrical connections, whether direct or indirect. Unless otherwise defined, features such as “parallel,” “perpendicular,” and “identical” used in the embodiments of the present application include strict cases of “parallel,” “perpendicular,” “identical,” as well as cases with a certain margin of error, such as “substantially parallel,” “substantially perpendicular,” “substantially identical.” For example, the term “substantially” may indicate a difference within 10% or within 5% of the average value of the objects being compared. When the quantity of a component or element is not specifically indicated in the subsequent description of the embodiments of the present application, it means that the component or element can be one or more, or at least one. “At least one” means one or more, and “multiple” means at least two.

[0032] Unless otherwise defined, all the technical and scientific terms used in this specification have the same meaning as understood by those skilled in the art to which the present application belongs. The terms used in this specification are only for the purpose of describing specific embodiments and are not intended to limit the present application. The term “and/or” as used herein includes any and all combinations of one or more of the associated listed items.

[0033] Furthermore, the technical features described in different embodiments of the present application can be combined with each other as long as they do not conflict. It is evident that the described embodiments are only a part of the embodiments of the present application, and not the entirety. All other embodiments that can be obtained by those skilled in the art without inventive labor based on the described embodiments of the present application are within the scope of protection of the present application.

[0034] FIG. 1 illustrates the method flowchart of the wig product production method according to an embodiment of the application. As shown in FIG. 1, the method includes:

[0035] **S100:** Preparing the first wig component.

[0036] The “first wig component” refers to a wig obtained through machine implanted techniques. The specific size and hairstyle can be adjusted according to the specific requirements and are not explicitly defined in this application.

[0037] **S200:** Dividing the elastic wig cap into the first region and the second region.

[0038] The elastic wig cap serves as the base for the wig that comes into contact with the wearer's head. The entire elastic wig cap can be divided into few regions to accommodate the first wig component and the second wig component, which are produced using different techniques. In this embodiment, for descriptive purposes, they are referred to as the “first region” and the “second region.” etc.

[0039] During practical implementation, the different regions of the elastic wig cap can be marked or distinguished using markers or other methods. The specific division of the elastic wig cap into regions can be determined based on specific requirements and is not explicitly defined.

[0040] In some embodiments, as shown in FIG. 10, the elastic wig cap can consist of the first elastic lace net **21** in the frontal area and the second elastic lace net **22** in the posterior area. The first elastic lace net **21** and the second elastic lace net **22** are stitched together using transparent

thread.

[0041] The first elastic lace net can be pre-shaped, while the second elastic lace net is unshaped. Lace net is a lightweight and transparent mesh material typically made from polymer fibers such as nylon or polyester, with a fine tie structure. It possesses lace-like texture and transparency, simulating the natural appearance of the scalp to provide a natural and realistic look for the wig product, allowing the wearer to confidently showcase different hairstyles and styles.

[0042] **S300**: Stitching and securing the first wig component to the first region of the elastic wig cap.

[0043] Stitching and securing can be achieved using various methods. For example, the first wig component can be stitched and secured to the elastic wig cap using transparent thread.

[0044] **S400**: Hand-tied the second wig component on the second region of the elastic wig cap.

[0045] **S500**: Trimming the transitional area between the first and second regions to generate the final wig product.

[0046] “Trimming” involves techniques such as manual crocheting, pruning, and combing to generate a more natural transition between the two different wig components and achieve the desired final wig product.

[0047] In practical use, the prepared wig is presented in the form of a wig cap, which can be directly worn on the wearer's head without requiring additional complicated steps, making it simple and convenient to use.

[0048] In some embodiments, as shown in FIG. 2, step **S100** described above can include:

[0049] **S110**: Fixing the hair bundles with a clamp.

[0050] The “hair bundles” refer to a bundle of hair formed by multiple strands of hair. Optionally, the hair strands can be made of synthetic fibers or human hair or mixture of both. When using synthetic fibers, the hair bundles can be fixed with a hair clamp after pruning. When using human hair, the same method can also be used.

[0051] **S120**: Creating a simulated scalp.

[0052] As shown in FIG. 3, the simulated scalp consists of a layered structure including a first scalp layer **11**, a second scalp layer **12**, and an elastic net **13**. The three layers can be fixed together using staples or similar methods.

[0053] Specifically, the first scalp layer is a simulated scalp made of PVC material, and the second scalp layer is a simulated scalp made of PU material. The thickness of the second scalp layer is significantly lower than that of the first scalp layer. This design provides the necessary support for machine implanted hair by the thicker first scalp layer, facilitating smooth hair implantation.

[0054] **S130**: Using a hair implantation machine to pass the root of the hair bundles through the simulated scalp.

[0055] “pass” refers to the root of the hair bundles passing through the first scalp layer of the simulated scalp, with the end of the hair bundles located on the outside of the elastic net, as shown in FIG. 3.

[0056] **S140**: Removing the first scalp layer of the simulated scalp.

[0057] “Removing” can be achieved by simply removing the staples, tearing it off, or similar methods, ensuring that the hair strands are not pulled out from the simulated scalp.

[0058] Specifically, before performing step **S140**, the following steps may be included:

[0059] Firstly, use a comb to arrange the roots of the hair bundles on the first surface of the simulated scalp. Then, prune the roots of the hair bundles to the predetermined length.

[0060] A “comb” is a tool used to comb and arrange hair, also known as a hairbrush or wide-tooth comb. It is typically made of plastic, wood, or metal and helps in the arrangement and combing of hair.

[0061] The first surface refers to the surface where the roots of the hair bundles emerge.

Accordingly, the predetermined length is the length of the hair strands protruding from the first scalp layer.

[0062] **S150**: Adhering and fixing the roots of the hair bundles onto the second scalp layer to generate the first wig component.

[0063] Before performing step **S150**, the roots of the hair bundles may be pruned again to the predetermined length, where the predetermined length is the length of the hair strands protruding from the second scalp layer.

[0064] In some embodiments, the predetermined length can be controlled to be 2 mm for better results.

[0065] Specifically, step **S150** can include the following steps:

[0066] Firstly, apply adhesive on the bonding surface of the second scalp layer and place it in an oven for baking. The bonding surface refers to the surface where the roots of the hair bundles protrude from the second scalp layer. After baking, reapply adhesive on the bonding surface of the second scalp layer and adhere the mesh layer. Finally, let the adhered mesh and simulated scalp air dry naturally to prepare the first wig component.

[0067] One advantageous aspect of the first wig component obtained through the above method is that by removing the first scalp layer, the thickness of the final wig product's scalp is effectively reduced, and there are no hair folds. This significantly improves the lightweight nature, making it approach the effect of hand-tied wigs.

[0068] In addition, different from the traditional machine-made wig product, as shown in FIG. 11, several holes are provided on the scalp of the first wig component. The scalp with holes may significantly improve the elasticity and breathability of the wig product.

[0069] To fully describe the production process of the wig product as described in this application, taking into account the specific examples shown in FIG. 4 to FIG. 10, the following is a detailed description of the specific preparation process of the wig according to this application:

[0070] 1) Preparation of the first wig component:

[0071] 1.1) Preparation of hair bundles:

[0072] taper multiple strands of hair on a hair clamp and trim or flatten the roots, then place the root portion of the hair strands under the head scalp needle. The hair ends of the hair bundles are fixed with a clamp.

[0073] 1.2) Preparation of simulated scalp:

[0074] Stack a regular PVC scalp, a PU scalp for hair implantation, and an elastic lace net in order, and fix them with staples. Then, cover the prepared hair bundles with the stacked layers.

[0075] Specifically, the regular PVC scalp is on top, the elastic lace net is below and in contact with the hair strands. Approximately 1 inch of hair roots is left on the top of the head.

[0076] 1.3) Hair implantation by machine:

[0077] Press the scalp crochet needle (model: 1-1 hair) through the scalp, hook the root of the hair strands through a hair hole, and only hook one hair strand with each stitch. As shown in FIG. 4, the upper layer of the PVC scalp 11 has short hair stops, and the elastic lace net below has long hair.

[0078] 1.4) Pruning:

[0079] As shown in FIG. 5, use a comb on the machine-implanted hair obtained in step 1.3) to comb the short hair stops on side A to side B. Prune the excessive hair stops on side B, leaving only 2 mm of hair roots.

[0080] 1.5) Removal of the regular PVC scalp:

[0081] As shown in FIG. 6, after pruning the hair stops, tear off the regular PVC scalp. It is important to control the force and speed while tearing to prevent the hair strands from being pulled out. After removing the regular PVC scalp, prune the hair stops again to leave only 2 mm of hair roots.

[0082] 1.6) Bonding roots and applying adhesive:

[0083] As shown in FIG. 7, after removing the PVC scalp, apply a thin layer of adhesive on the side with short hair stops and place it in the oven to partially dry. Then, as shown in FIG. 8, apply another thin layer of adhesive and attach a breathable mesh 14. Let it air dry naturally to obtain the

desired first wig component.

[0084] 2) Preparation of the wig product:

[0085] 2.1) Taper the hair strands on a hair clamp.

[0086] 2.2) As shown in FIG. 9, use a pre-shaped curved first elastic lace net **21** for the front and an unshaped second elastic lace net **22** for the back. Stitch the first elastic lace net and the unshaped second elastic lace net together with transparent fishing line to generate an elastic wig cap.

[0087] 2.3) Mark the first region and the second region on the elastic wig cap using a marker pen.

[0088] 2.4) Use transparent fishing line to stitch the first wig component onto the marked first region.

[0089] 2.5) As shown in FIG. 10, use a manual crochet method to hook the hair strands onto the mesh holes in the second region.

[0090] 2.6) Use a manual crochet method to generate a natural transition between the first wig component and the second wig component in a transition area.

[0091] 2.7) Further trim and comb the finished product obtained in step 2.6 to obtain the wig presented in the form of a wig cap.

[0092] The present application further provides a wig producing according to above producing method. The wig is consist of the machine implantation part (ie, the first wig component) and the hand-tied part (ie, the second wig component).

[0093] The above content provides further detailed explanations in conjunction with specific/preferred embodiments, but it should not be construed that the specific embodiments of the present application are limited to these explanations. Those skilled in the art can make various modifications and improvements within the scope of the present application without departing from the spirit of the present application.

## Claims

1. A method for producing wig, comprising: preparing a first wig component obtained through a machine hair implantation process; dividing an elastic wig cap into a first region and a second region; stitching and securing the first wig component to the first region of the elastic wig cap; hand-tied a second wig component on the second region of the elastic wig cap; trimming a transitional area between the first region and the second region to generate a wig.

2. The method according to claim 1, wherein the step of preparing a first wig component obtained through a machine hair implantation process comprises: fixing hair bundles with a clamp; wherein the hair bundles are formed by converging multiple hair strands; creating a simulated scalp consists of a layered arrangement of a first scalp layer, a second scalp layer, and an elastic mesh; using a hair implantation machine to pass a root of the hair bundles through the simulated scalp; wherein the roots of the hair bundles emerge from the first scalp layer of the simulated scalp, while the ends of the hair bundles are positioned on the outer side of the elastic mesh; removing the first scalp layer of the simulated scalp; adhering and securing the roots of the hair bundles onto the second scalp layer to generate the first wig component.

3. The method according to claim 2, wherein before the step of removing the first scalp layer of the simulated scalp, the method further comprises: using a wide-toothed comb to arrange the roots of the hair bundles on the first surface of the simulated scalp; pruning the roots of the hair bundles to a predetermined length.

4. The method according to claim 2, wherein before the step of adhering and securing the roots of the hair bundles onto the second scalp layer, the method further comprises: pruning the roots of the hair bundles again to the predetermined length; wherein the predetermined length is the length at which the hair strands protrude from the second scalp layer.

5. The method according to claim 4, wherein the predetermined length is 2 mm.

6. The method according to claim 2, wherein the first scalp layer is a simulated scalp made of PVC

material, and the second scalp layer is a simulated scalp made of PU material, with the second scalp layer significantly thinner than the first scalp layer.

**7.** The method according to claim 2, wherein the step of adhering and securing the roots of the hair bundles onto the second scalp layer to generate the first wig component comprises: coating adhesive on the mating surface of the second scalp layer and placing it in an oven for baking; wherein the mating surface is the surface where the roots of the hair bundles protrude from the second scalp layer; after baking, reapplying adhesive on the mating surface of the second scalp layer and attaching a mesh layer; allowing the attached mesh layer on the simulated scalp to air dry naturally to prepare the first wig component.

**8.** The method according to claim 1, wherein the first wig component is stitched and secured to the elastic wig cap using a transparent thread.

**9.** The method according to claim 2, wherein the hair strands are made of synthetic fiber, human hair or mixture of both.

**10.** The method according to claim 1, wherein the frontal part of the elastic wig cap is a curved first elastic lace net, and the remaining part of the elastic wig cap is an unshaped second elastic lace net; wherein the first elastic lace net and the second elastic lace net are stitched together using transparent thread.

**11.** A wig obtained by a method for producing wig; wherein the method comprising: preparing a first wig component obtained through a machine hair implantation process; dividing an elastic wig cap into a first region and a second region; stitching and securing the first wig component to the first region of the elastic wig cap; hand-tied a second wig component on the second region of the elastic wig cap; trimming a transitional area between the first region and the second region to generate a wig.

**12.** The wig according to claim 11, wherein the step of preparing a first wig component obtained through a machine hair implantation process comprises: fixing hair bundles with a clamp; wherein the hair bundles are formed by converging multiple hair strands; creating a simulated scalp consists of a layered arrangement of a first scalp layer, a second scalp layer, and an elastic mesh; using a hair implantation machine to pass a root of the hair bundles through the simulated scalp; wherein the roots of the hair bundles emerge from the first scalp layer of the simulated scalp, while the ends of the hair bundles are positioned on the outer side of the elastic mesh; removing the first scalp layer of the simulated scalp; adhering and securing the roots of the hair bundles onto the second scalp layer to generate the first wig component.

**13.** The wig according to claim 12, wherein before the step of removing the first scalp layer of the simulated scalp, the method further comprises: using a wide-toothed comb to arrange the roots of the hair bundles on the first surface of the simulated scalp; pruning the roots of the hair bundles to a predetermined length.

**14.** The wig according to claim 12, wherein before the step of adhering and securing the roots of the hair bundles onto the second scalp layer, the method further comprises: pruning the roots of the hair bundles again to the predetermined length; wherein the predetermined length is the length at which the hair strands protrude from the second scalp layer.

**15.** The wig according to claim 14, wherein the predetermined length is 2 mm.

**16.** The wig according to claim 12, wherein the first scalp layer is a simulated scalp made of PVC material, and the second scalp layer is a simulated scalp made of PU material, with the second scalp layer significantly thinner than the first scalp layer.

**17.** The wig according to claim 12, wherein the step of adhering and securing the roots of the hair bundles onto the second scalp layer to generate the first wig component comprises: coating adhesive on the mating surface of the second scalp layer and placing it in an oven for baking; wherein the mating surface is the surface where the roots of the hair bundles protrude from the second scalp layer; after baking, reapplying adhesive on the mating surface of the second scalp layer and attaching a mesh layer; allowing the attached mesh layer on the simulated scalp to air dry

naturally to prepare the first wig component.

**18.** The wig according to claim 11, wherein the first wig component is stitched and secured to the elastic wig cap using a transparent thread.

**19.** The wig according to claim 12, wherein the hair strands are made of synthetic fiber, human hair or mixture of both.

**20.** The wig according to claim 11, wherein the frontal part of the elastic wig cap is a curved first elastic lace net, and the remaining part of the elastic wig cap is an unshaped second elastic lace net; wherein the first elastic lace net and the second elastic lace net are stitched together using transparent thread.

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